Volumes of Revolution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The disk and washer methods**

Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.

$$V = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\tfrac{x^2}{2}\right]_0^4 = 8\pi.$$

2 **[Foundational]** • **Target: The disk and washer methods**

The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

Bounds $x = -2$ to 2: $V = \pi\int_{-2}^{2}[4^2 - (x^2)^2]dx = \pi\int_{-2}^{2}(16 - x^4)\,dx = \pi\left[16x - \tfrac{x^5}{5}\right]_{-2}^{2} = \tfrac{384\pi}{5}$.

3 **[Foundational]** • **Target: The shell method**

The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.

$$V = 2\pi\int_0^2 x \cdot x^2\,dx = 2\pi\int_0^2 x^3\,dx = 2\pi\left[\tfrac{x^4}{4}\right]_0^2 = 8\pi.$$

4 **[Foundational]** • **Target: The shell method**

The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.

$$V = 2\pi\int_0^3 x \cdot x\,dx = 2\pi\left[\tfrac{x^3}{3}\right]_0^3 = 18\pi.$$

5 **[Foundational]** • **Target: Comparing methods**

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis (not the y-axis). Find the volume using the disk method.

$$V = \pi\int_0^2 (x^2)^2\,dx = \pi\int_0^2 x^4\,dx = \pi\left[\tfrac{x^5}{5}\right]_0^2 = \tfrac{32\pi}{5}.$$

6 **[Foundational]** • **Target: Comparing methods**

The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x-axis. Find the volume using the washer method.

Intersections: $\sqrt{x} = x$ at $x = 0, 1$. On $[0, 1]$, $\sqrt{x} \geq x$, so the outer radius is $\sqrt{x}$, inner is x:
$V = \pi\int_0^1 [(\sqrt{x})^2 - x^2]dx = \pi\int_0^1 (x - x^2)\,dx = \pi\left[\tfrac{x^2}{2} - \tfrac{x^3}{3}\right]_0^1 = \tfrac{\pi}{6}$.

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7 [Foundational] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.

$V = \pi\int_0^2 [(x^2+1)^2 - 1^2]\,dx = \pi\int_0^2 (x^4 + 2x^2)\,dx = \pi\left[\frac{x^5}{5} + \frac{2x^3}{3}\right]_0^2 = \pi\left(\frac{32}{5} + \frac{16}{3}\right) = \frac{176\pi}{15}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).

$V = 2\pi\int_0^2 (3-x)x^2\,dx = 2\pi\int_0^2 (3x^2 - x^3)\,dx = 2\pi\left[x^3 - \frac{x^4}{4}\right]_0^2 = 2\pi(8-4) = 8\pi$.

9 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis. Find the volume, using $\sin^2 x = \frac{1 - \cos 2x}{2}$.

$V = \pi\int_0^\pi \sin^2 x\,dx = \pi\int_0^\pi \frac{1-\cos 2x}{2}\,dx = \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^\pi = \frac{\pi}{2}(\pi - 0) = \frac{\pi^2}{2}$.

10 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x-axis. Find the volume, to three decimal places.

$V = \pi\int_0^1 e^{-2x}\,dx = \pi\left[-\frac{1}{2}e^{-2x}\right]_0^1 = \frac{\pi}{2}(1 - e^{-2}) \approx \frac{\pi}{2}(0.8647) \approx 1.358$.

11 [Intermediate] • Target: Setting up without evaluating

Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x-axis.

$V = \pi\int_1^e (\ln x)^2\,dx$.

12 [Intermediate] • Target: Setting up without evaluating

Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y-axis, using the shell method.

$V = 2\pi\int_0^1 x(\sqrt{x} - x^2)dx$ (outer curve $\sqrt{x}$ minus inner curve x^2, since $\sqrt{x} \geq x^2$ on $[0, 1]$).

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13 **[Intermediate]** • **Target: Visualizing the generating region**

The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x-axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.

The solid is a paraboloid-like horn, narrow at $x = 0$ and widening to radius 2 at $x = 4$. A full cylinder of radius 2, length 4 would have volume $\pi(2)^2(4) = 16\pi$; since the paraboloid is narrower than the cylinder everywhere except at $x = 4$, a volume of 8π (exactly half the cylinder) is reasonable.

14 **[Intermediate]** • **Target: Volume of a solid with a hole (washer method practice)**

The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x-axis. Find the volume using the washer method (outer radius x, inner radius x^3).

$V = \pi\int_0^1 [x^2 - x^6]dx = \pi\left[\frac{x^3}{3} - \frac{x^7}{7}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{7}\right) = \frac{4\pi}{21}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: An applied volume problem** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x-axis, where x is the height (cm) and y is the radius. Find the volume of the vase.

$V = \pi\int_0^4 (\sqrt{x} + 1)^2\,dx = \pi\int_0^4 (x + 2\sqrt{x} + 1)dx = \pi\left[\frac{x^2}{2} + \frac{4}{3}x^{3/2} + x\right]_0^4 = \pi\left(8 + \frac{32}{3} + 4\right) = \pi\left(\frac{68}{3}\right) = \frac{68\pi}{3}$ cm^3.

16 **[Challenging]** • **Target: Volume with the shell method, revisited** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y-axis. Use the shell method to find the volume.

$V = 2\pi\int_0^2 x(4 - x^2)\,dx = 2\pi\int_0^2 (4x - x^3)\,dx = 2\pi\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2\pi(8 - 4) = 8\pi$.

17 **[Challenging]** • **Target: Visualizing the generating region** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

Apply the volume of revolution formula: $V = \pi\int_0^4 y^2\,dx = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = \pi\left(\frac{16}{2} - 0\right) = 8\pi$.

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18 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

1. Step formula: $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h(x_n + y_n)$ with $h = 0.1$, $x_0 = 0$, $y_0 = 1$. Step 1 (to $x_1 = 0.1$): $y_1 = y_0 + 0.1(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.1$. Step 2 (to $x_2 = 0.2$): $y_2 = y_1 + 0.1(x_1 + y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.1(1.2) = 1.1 + 0.12 = 1.22$. Euler's approximation is $y(0.2) \approx 1.22$. 2. Exact analytical solution: Rewrite as $\frac{dy}{dx} - y = x$. Integrating factor $\mu = e^{-x}$.

19 [Challenging] • Target: An applied volume problem • Cross-Lesson: 6.01 Related Rates

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 6.02 Optimization and Curve Sketching

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The disk and washer methods

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$

a. $f(x)$

b. $b - a$

c. a

d. x ✓

Applying Applications of Integration principles, Shell radius $= x$. Therefore, the verified correct option is x.

22 [Foundational] • Target: The shell method

Volume of solid of revolution of $y = \cos x$ about x-axis from 0 to $\pi/2$:

a. $\pi^2/2$

b. $\pi/2$

c. $\pi^2/4$ ✓

d. $\pi/4$

$\pi\int_0^{\pi/2} \cos^2 x\, dx = \pi \cdot \pi/4 = \pi^2/4$.

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23 [Foundational] • Target: Comparing methods

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. **They give the same volume ✓**

c. Washer is always larger

d. Depends on axis and region

Both methods give the same volume for the same solid.

24 [Foundational] • Target: Volumes about other axes

Which gives volume by the disk method about the y-axis?

a. $\int_a^b f(y)\,dy$

b. $\pi\int_a^b [f(x)]^2\,dx$

c. $2\pi\int_a^b xf(x)\,dx$

d. $\pi\int_c^d [f(y)]^2\,dy$ **(solve for x first) ✓**

Rotate about y-axis; use $x = f(y)$.

25 [Foundational] • Target: Volumes with trigonometric and exponential curves

Volume of the solid of revolution of $y = \sqrt{9 - x^2}$ (semicircle) about x-axis:

a. 18π

b. $\frac{4}{3}\pi \cdot 27 = 36\pi$

c. 36π ✓

d. $\frac{4}{3}\pi(3)^3 = 36\pi$

$V = \pi\int_{-3}^{3}(9 - x^2)dx = 36\pi.$

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26 [Foundational] • Target: Setting up without evaluating

A region R is bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi \ln(10)$

b. $2\pi \ln(5)$

c. $\pi \ln(5)$ ✓

d. $2\pi \ln(10)$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and $x = a, x = b$ about the y-axis can be found using the shell method. The formula for the volume is $V = \int_a^b 2\pi x f(x)\,dx$. \\ In this case, $f(x) = \frac{1}{x^2+1}$, $a = 1$, and $b = 3$. \\ So, the volume is $V = \int_1^3 2\pi x\left(\frac{1}{x^2+1}\right)dx$. \\ We can pull out the constant 2π: $V = 2\pi\int_1^3 \frac{x}{x^2+1}\,dx$. \\ To evaluate this integral, we use a substituti.

27 [Foundational] • Target: Visualizing the generating region

Volume by shell about x-axis: shells have radius $=$

a. y ✓

b. x

c. $f(x)$

d. $f(y)$

Applying Applications of Integration principles, Shell about x-axis: radius $= y$. Therefore, the verified correct option is y.

28 [Foundational] • Target: Volume of a solid with a hole (washer method practice)

The region under $y = \sin(x)$ for $0 \le x \le \pi$ is revolved about the x-axis. Find the volume.

a. $\frac{\pi}{2}$

b. π^2

c. $\frac{\pi^2}{2}$ ✓

d. $2\pi^2$

Disk method: $V = \pi\int_0^\pi \sin^2(x)\,dx = \frac{\pi^2}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: An applied volume problem

Volume by disc method rotating $y = \sqrt{x}$ about x-axis from 0 to 4:

a. 4π

b. 16π

c. 8π ✓

d. 32π

$V = \pi\int_0^4 (\sqrt{x})^2 dx = \pi[x^2/2]_0^4 = 8\pi$.

30 [Intermediate] • Target: Volume with the shell method, revisited

Volume of the region $0 \le x \le 1$, $0 \le y \le e^x$ rotated about $y = -1$:

a. $\pi(e^2 - 2e + 1)/2$

b. $\pi(e^2 - 2e)/2$

c. $\pi(e^2 - 3)/2$ ✓

d. $2\pi(e - 2)$

Washer: $\pi\int_0^1 ((e^x + 1)^2 - 1)dx$.

31 [Intermediate] • Target: The disk and washer methods

Volume of the solid obtained rotating $y = \sin x$ $(0 \le x \le \pi)$ about the x-axis:

a. $\pi^2/4$

b. $\pi^2/2$ ✓

c. $\pi/2$

d. π

$V = \pi\int_0^\pi \sin^2 x\, dx = \pi^2/2$.

32 [Intermediate] • Target: The shell method

Pappus' theorem: solid of revolution of right triangle $(0,0), (1,0), (0,1)$ about y-axis: $\bar{x} = 1/3$, $A = 1/2$:

a. $\pi/2$

b. $\pi/6$

c. $\pi/3$ ✓

d. $2\pi/3$

$V = 2\pi \cdot (1/3) \cdot (1/2) = \pi/3$.

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33 [Intermediate] • Target: Comparing methods

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$ ✓

d. $\frac{38}{5}\pi$

Washer method: $V = \pi\int_0^1 (2^2 - (x^2)^2)dx = \frac{19}{5}\pi$.

34 [Intermediate] • Target: Volumes about other axes

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$ ✓

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^2 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$.

35 [Intermediate] • Target: Volumes with trigonometric and exponential curves

Volume rotating $y = 2$ about the x-axis from 0 to 3:

a. 4π

b. $4\pi \cdot 3 = 12\pi$

c. 6π

d. 12π ✓

Applying Applications of Integration principles, $V = \pi\int_0^3 4\,dx = 12\pi$. Therefore, the verified correct option is 12π.

36 [Intermediate] • Target: Setting up without evaluating

Moment of inertia of spherical shell $x^2 + y^2 + z^2 = R^2$ about z-axis using surface integral:

a. $\frac{2}{5}MR^2$

b. $\frac{2}{3}MR^2$ ✓

c. MR^2

d. $\frac{1}{2}MR^2$

Applying Vector Calculus principles, Shell: $I_z = \frac{2}{3}MR^2$. Therefore, the verified correct option is $\frac{2}{3}MR^2$.

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37 [Intermediate] • Target: Visualizing the generating region

Volume of spherical shell $1 \le \rho \le 2$:

a. 4π

b. $28\pi/3$ ✓

c. 8π

d. $4\pi/3$

Applying Multiple Integrals principles, $V = (4\pi/3)(8-1) = 28\pi/3$. Therefore, the verified correct option is $28\pi/3$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

For a region symmetric about the y-axis, $\iint_D x\,dA =$

a. 0 ✓

b. -1

c. 1

d. $A(D)$

Applying Multiple Integrals principles, By symmetry: $\iint x\,dA = 0$. Therefore, the verified correct option is 0.

39 [Challenging] • Target: An applied volume problem • Cross-Lesson: 6.01 Related Rates

Volume of solid V using divergence theorem: $V = \frac{1}{3}\oiint_S \mathbf{r}\cdot d\mathbf{S}$ since $abla \cdot \mathbf{r} =$

a. 3 ✓

b. 1

c. 0

d. $\mathbf{r}$

Applying Vector Calculus principles, $abla \cdot (x, y, z) = 3$. Therefore, the verified correct option is 3.

40 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 6.02 Optimization and Curve Sketching

Find the volume of the solid between the paraboloid $z = x^2 + y^2$ and the plane $z = 1$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$ ✓

d. $\frac{3\pi}{2}$

In cylindrical coordinates, $V = \int_0^{2\pi}\int_0^{\sqrt{1}} (1-r^2)\,r\,dr\,d\theta = \frac{\pi}{2}$.

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41 [Challenging] • Target: The disk and washer methods • Cross-Lesson: 7.01 Integration by Substitution

Volume of region $0 \le z \le 1 - x^2 - y^2$ (positive):

a. $\pi/4$

b. $\pi/2$ ✓

c. π

d. $3\pi/4$

$V = \int_0^{2\pi}\int_0^1 (1 - r^2) r\, dr\, d\theta = 2\pi \cdot [r^2/2 - r^4/4]_0^1 = 2\pi \cdot 1/4 = \pi/2.$

42 [Challenging] • Target: The shell method • Cross-Lesson: 7.02 Integration by Parts

Find the volume of the solid under $z = 3x + y$ above the unit square $[0,1] \times [0,1]$.

a. 2 ✓

b. 4

c. 1

d. 6

$\int_0^1\int_0^1\int_0^{3x+y} 1\, dz\, dy\, dx \;=\; 2.$

43 [Challenging] • Target: Comparing methods • Cross-Lesson: 8.01 Area Between Curves

Kelvin-Stokes theorem applied to $\mathbf{F} = (y, 0, 0)$ over disk $z = 0, x^2 + y^2 \le 1$:

a. 2π

b. π

c. 0

d. $-\pi$ ✓

$abla \times \mathbf{F} = (0, 0, -1)$; flux $= -\pi R^2 = -\pi.$

44 [Challenging] • Target: Volumes about other axes • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Volume by shells revolving $y = x^2$ about the y-axis from 0 to 1:

a. $\pi/2$ ✓

b. $\pi/4$

c. π

d. $2\pi/4$

$V = 2\pi\int_0^1 x \cdot x^2\, dx = 2\pi[x^4/4]_0^1 = \pi/2.$

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45 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 9.01 Separable Differential Equations

Volume of solid with square cross-sections perpendicular to x-axis, bounded by $y = \sqrt{x}$ and $x = 4$:

a. 4

b. 8 ✓

c. 16/3

d. 8/3

$A(x) = (\sqrt{x})^2 = x$; $V = \int_0^4 x\,dx = 8$.

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